ENVIRONMENTAL INTELLIGENCE × AIR QUALITY

CONTINUOUS AMBIENT AIR QUALITY MONITORING

Sensor-Based *Ambient* Air Quality Monitor

Gases, particulate matter and weather from one solar-powered outdoor station — measured continuously and streamed to the cloud every minute.

PM1 · 2.5 · 10

LASER PARTICULATE CHANNELS

6 gases

SELECTABLE FROM 10-GAS LIBRARY

60 sec

CLOUD UPLOAD INTERVAL

SOLAR POWERED OFF-GRID READY | **IP 54** OUTDOOR ENCLOSURE | **<3.5 KG** POLE MOUNTED

16	15s	4s	1.2W
POLLUTANT & WEATHER CHANNELS	GAS RESPONSE TIME	PARTICLE RESPONSE TIME	PEAK POWER CONSUMPTION

01 OVERVIEW

A full monitoring station, *on a single pole.*

A sensor-based continuous ambient air quality monitoring station that delivers real-time, precise and reliable data for industrial, urban and environmental applications — without the footprint, shelter or utilities of a conventional analyzer cabin.

Advanced electrochemical gas cells and optical laser particulate sensing sit alongside a full weather station in one compact, weather-resistant housing. Low power draw and an efficient solar system let the station run off-grid, while cloud connectivity turns every reading into an accessible, alertable data point.

MEASURED CHANNELS

- PM1
- PM2.5
- PM10
- CO
- CO_2
- NO_2
- SO_2
- O_3
- H_2S
- NH_3
- CH_4
- TVOC
- HCL
- TEMP
- RH
- PRESSURE
- WIND SPEED
- WIND DIRECTION
- SOLAR RADIATION

· PRODUCT FEATURES

Real-Time Multi-Gas Detection

High-precision electrochemical cells measure any six gases from the library, with 15–30 second response.

Optical Laser Particulates

PM1, PM2.5 and PM10 resolved simultaneously by laser scattering, updating in as little as 4 seconds.

Solar Power & Low Draw

0.08–1.2 W consumption with a 6 Ah lithium pack gives 13–14 hours of autonomous backup.

Cloud & IoT Integration

Data to the cloud every minute over Ethernet, RS-485 or 2G/3G/4G, with remote access and alerts.

02 MONITORED PARAMETERS

Pollutants, particles & *meteorology.*

ELECTROCHEMICAL GAS SENSORS — ANY SIX SELECTED

Carbon Monoxide CO	ppb / ppm
Nitrogen Dioxide NO_2	ppb
Ozone O_3	ppb
Ammonia NH_3	ppb / ppm
Total VOCs TVOC	ppb
Carbon Dioxide CO_2	ppm
Sulphur Dioxide SO_2	ppb
Hydrogen Sulphide H_2S	ppb
Methane CH_4	ppm
Hydrogen Chloride HCl	ppm

OPTICAL LASER — PARTICULATE MATTER — STANDARD FITMENT

Particulate Matter PM1	µg/m³
Coarse Particulate PM10	µg/m³
Fine Particulate PM2.5	µg/m³

INTEGRATED WEATHER STATION

Ambient Temperature	-10 … 60 °C
Barometric Pressure	hPa
Wind Direction	0 … 360°
Relative Humidity	0 … 99 %RH
Wind Speed	m/s
Solar Radiation	W/m²

Gas configuration is fixed at order — any six sensors from the library are fitted in the standard housing. Additional channels available on request.

03 HOW IT WORKS

Sensing, weather & *data path.*

Two independent sensing technologies cover gases and particles, a meteorological head puts every reading in context, and an IoT data path carries results from the pole to your dashboard.

ELECTROCHEMICAL

Catalytic Reaction

CO · CO_2 · NO_2 · SO_2 · O_3 · H_2S · NH_3 · CH_4 · TVOC

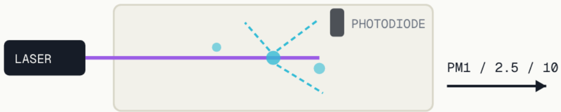
Sample air diffuses into the sensing cell where the target gas reacts at the working electrode. The resulting current is proportional to concentration and is linearised against calibration.

`Response 15-30 s`

OPTICAL LASER SCATTERING

Particle Size Classification

PM1 · PM2.5 · PM10



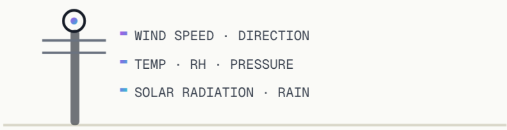
A laser beam illuminates the sample stream; particles scatter light onto a photodiode. Pulse amplitude and count give both size distribution and mass concentration.

`Response from 4 s`

METEOROLOGY

Integrated Weather Head

WIND · TEMP · RH · PRESSURE · SOLAR



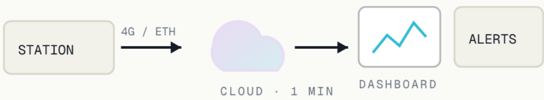
Co-located meteorological sensing lets pollutant readings be interpreted against wind, temperature and radiation — essential for source apportionment and dispersion assessment.

`Same time base as pollutants`

CONNECTIVITY

IoT Data Path

ETHERNET · RS-485 · 2G / 3G / 4G



Measurements are pushed to the cloud once every minute for visualisation, trending and threshold alerts. MODBUS RTU slave output integrates the station with local SCADA.

`Remote management available`

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

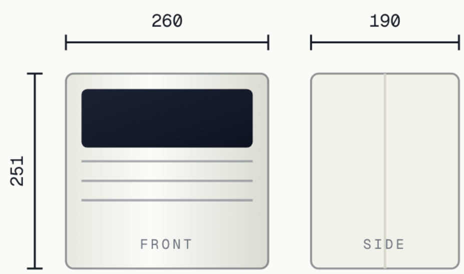
Gas Sensors	Any six of CO, CO_2, NO_2, O_3, SO_2, H_2S, NH_3, CH_4, TVOC, HCl
PM Sensor	PM1, PM2.5 & PM10
Working Principle	Electrochemical (gases) · optical laser (PM)
Weather Station	Temp, RH, pressure, wind speed & direction, solar radiation
Response — Gases	15 - 30 seconds
Response — Particles	From 4 seconds
Operating Temperature	-10 - 60 °C
Operating Humidity	0 - 99 %RH
Ingress Protection	IP 54

POWER · COMMS · BUILD

Battery	Lithium 6 Ah · 13-14 h backup
External Supply	12-18 VDC · charger or solar panel
Power Consumption	0.08 - 1.2 W by configuration
Communication	Ethernet · RS-485 · 2G / 3G / 4G SIM
Connectors	Power 12-18 V · MODBUS RTU slave · weather station · Ethernet
Cloud Upload	Every 1 minute
Remote Management	Supported
Dimensions (L×W×H)	260 × 190 × 251 mm
Weight	< 3.5 kg
Enclosure	Stainless steel & aluminium

ENCLOSURE DIMENSIONS

FRONT & SIDE ELEVATION · MM



05 SITING & INSTALLATION

Placed in the *breathing zone.*

As per CPCB guidance for continuous ambient air quality monitoring stations, the unit is mounted at approximately **2 m above ground level**, with the air inlet positioned **between 2 and 4 m** from the ground — so measurements represent the air people actually breathe.

01 · MOUNTING

Single pole or wall

Under 3.5 kg, the station clamps to a standard pole or wall bracket. No shelter, foundation or air-conditioned cabin required.

02 · INLET HEIGHT

2 – 4 m from ground

Inlet kept clear of obstructions, walls and vegetation so the sample represents open ambient air at breathing height.

03 · POWER

Solar or 12–18 VDC

Efficient solar system with a 6 Ah lithium pack carries the station through the night and through grid outages.

04 · CONNECTIVITY

SIM, Ethernet or RS-485

Cellular SIM for remote sites; Ethernet or MODBUS RTU where a plant network or SCADA is already present.

05 · COMMISSIONING

Calibrated and ready

Sensors arrive calibrated and configured. On-site setup is bracket, power, SIM and a connectivity check.

06 · UPKEEP

Low maintenance

No reagents or consumables. Periodic sensor verification and inlet cleaning; remote diagnostics flag drift early.

NETWORK SCALABILITY

Because each station is self-contained and low-power, a monitoring network can start with a single unit and grow to a city-wide grid — every station reporting to the same cloud platform, with a consistent time base and a single dashboard for comparison across sites.

06 APPLICATIONS

Where it *works.*

The same station serves very different sites — a plant boundary, a city junction, a construction perimeter or a remote off-grid location — because it needs only power and a network signal.

Industrial Plant Boundaries	Urban & City Networks	Roadside & Traffic Corridors
Construction Sites	Mining & Quarrying	Chemical & Pharma Campuses
Ports, Yards & Logistics Hubs	Power & Thermal Stations	Remote & Off-Grid Sites

TYPICAL DEPLOYMENTS

Most installations fall into one of two patterns: a boundary ring of two to four stations around a single site, reporting compliance-grade trends at the fence line; or a distributed grid across a city or industrial cluster, where spatial coverage matters more than any one station. Both run on the same hardware, the same cloud platform and the same reporting.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Sensor-based ambient monitoring: electrochemical catalytic cells for gases and optical laser scattering for particulate matter, with a co-located weather station.

Q Which parameters can it measure?

PM1, PM2.5 and PM10 plus gaseous pollutants — CO, CO_2, NO_2, SO_2, O_3, NH_3, H_2S, CH_4, TVOC and HCl — any six gases configured to your site requirement.

Q Is it suitable for continuous monitoring?

Yes. It is designed for real-time, continuous operation in industrial, urban and environmental applications, uploading to the cloud every minute.

Q Can it be installed outdoors?

Yes. The stainless steel and aluminium housing is IP 54 rated and weather resistant, operating from −10 to 60 °C and 0–99 %RH.

Q How is it powered at remote sites?

An efficient solar system with a 6 Ah lithium battery gives 13–14 hours of backup; a 12–18 VDC charger can be used where mains is available.

Q Does it support cloud and remote access?

Yes — IoT-enabled cloud monitoring with real-time data access, visualisation, customisable alerts and remote station management.

Q How does it integrate with SCADA?

A dedicated MODBUS RTU slave connector is provided, alongside Ethernet and RS-485, so the station can report into existing plant systems.

Q At what height should it be mounted?

Around 2 m above ground level with the inlet between 2 and 4 m, per CPCB guidance, so readings represent the breathing zone.

Q How fast are the readings?

Gas channels respond in 15–30 seconds and particulate channels from 4 seconds, giving a genuinely live picture of site air quality.

Q Can a network be expanded later?

Yes. Stations are self-contained and scalable — add units to the same cloud platform to build a multi-point or city-wide network.

MONITORING AIR, IN REAL TIME

Know your air.
Continuously.

Tell us your site and the pollutants you need to report on, and we'll configure the station — gases, particulates, weather, power and connectivity to match. Pick how you'd like to start.

01

Talk to our engineers

Speak directly to the team that specifies and commissions ambient monitoring stations. No call centre.

+91 99538 86178 →

02

Request a station configuration

RECOMMENDED

Send us your site and reporting requirement — we'll spec the gas set, PM channels, weather head and power.

sales@vepolink.com →

03

Book a demonstration

See a live station and the cloud dashboard with our engineering team.

Request a demo →